Symmetric Key Encryption

- Stream cipher (Encrypt or Decrypt bit wise)
- Block Cipher (n bit block wise)
  - 64 bit
  - 128 bit
  - 256, 192 bit

$128 \text{ bit} = 2^7 \text{ bit}$

Kilo = 1000 or 2^{10}

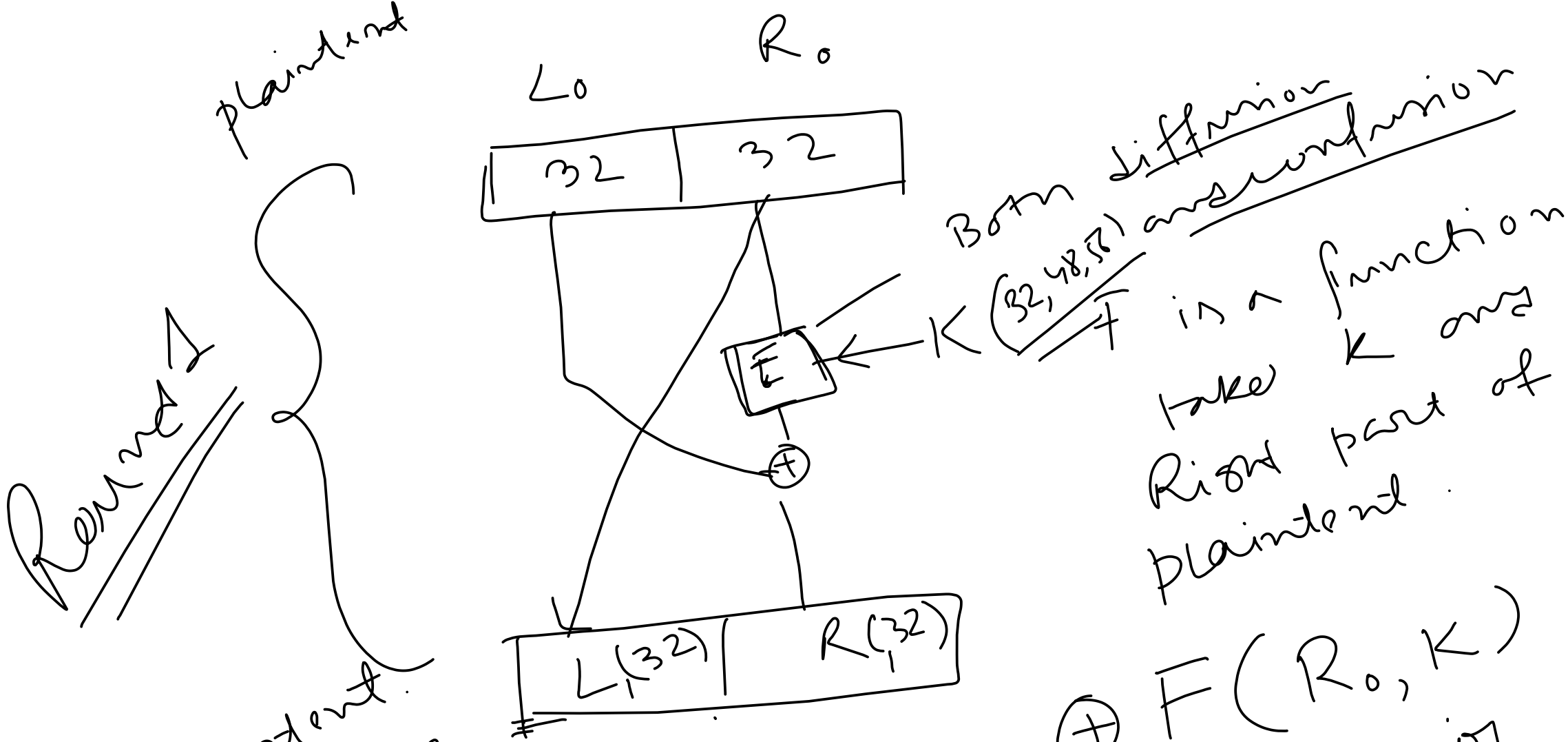
2 MB of data.

$2 \times 1024 \text{ KB} = 2 \times 1024 \times 1024 \text{ B}$

$= 2 \times 1024 \times 1024 \times 8 \text{ bit}$

$= 2^1 \times 2^{10} \times 2^{10} \times 2^3 = 2^{24} \text{ bit}$

We need $2^{24} / 2^7 = 2^{17}$ blocks to encrypt.



f is a function take K and Right part of plaintext.

multiple times running in 16 times

$$R_1 = L_0 \oplus F(R_0, K)$$

$$L_1 = R_0$$

(P|C) pairs

Exhaustive search $= 2^{56}, 2^{64}$

64 bit key

Double DES

$C = E_K(m)$

$C = E_{K_1}\left(E_{K-2}(m)\right)$

16 Rounds

1/2 Rounds we have direct attack

Chosen Plaintext Attack (CPA)

Adversary — challenger

m_0, m_1 →

$|m_0| = |m_1|$

← $b \xleftarrow{R} \{0,1\}$

← $c = E_K(m_b)$

find b

$b' = b$ wins

Indistinguishability Game

Adversary can query with some messages and get some ciphertext.

m_0, m_1 get c_0 c_1

$c = E_{K_1}[F_{K_2}(m)]$ $K_1 \| K_2$

Goal of Adversary to find the K_1 and K_2.

2^{56}

DFS

2^{112}

$C = E_{K_1}\left[E_{K_2}(m)\right]$

Under all possible K find.

$E_K(m) = C_1, K_1'$
C_2, K_2'
$\vdots$
$C_{2^{56}}, K_{2^{56}}'$

$E_K^{-1}(c) = m_1', K_1'$
m_2, K_2'
$\vdots$
$m_{2^{56}}, K_{2^{56}}'$

$K_2' = K_\beta'$

K_1 K_2

2^{56}

$2^{56} - 2^{56} + 2 = 2^{57}$

2^{56}

K_1 112 bit

2^{56}

57 bit

Triple DES

$$C = E_{K_1}\left(E_{K_2}\left(E_{K_1}(m)\right)\right) \quad 3, K_1$$

two different keys

NOT used by US NSA

AES arrived

DES was discontinued.

(1) (2)

SPN

Block Cipher

64 — DES runs — Diffusion
AES — confusion

128, 192, 256

permutation
Shifting
expansion
compression
} multiple times

Meet in the middle attack (Double DES) 1971

2001 AES

1 rounds attack.

2 rounds

i/p input is

$O^{32}(0, \ldots 1)$

$R = 11 \ldots$

16 rounds